07/07/2021

Pledgement: "On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

1507 20056

Question 1

$$A=\begin{bmatrix} 1 & x & -y & x \\ 3x & 5 & -9y & 1 \\ 2 & B & 3 & x^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \xrightarrow[r_4 - 1\,r_1]{r_2 - 3x\,r_1,\ r_3 - 2\,r_1} \begin{bmatrix} 1 & x & -y & x \\ 0 & 5-3x^2-9y+3xy & & 1-3x^2 \\ 0 & B-2x & 3+2y & x^2-2x \\ 0 & 14-x & y-6 & 2-x \end{bmatrix}$$

$$A = x\begin{bmatrix} 0 & 1 & 0 & 1 \\ 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & x \\ 0 & 0 & 0 & 0 \end{bmatrix} + y\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & -9 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + B\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & x & -y & x \\ 0 & 1 & \frac{3xy-9y}{5-3x^2} & \frac{1-3x^2}{5-3x^2} \\ 0 & B-2x & 3+2y & x^2-2x \\ 0 & 14-x & y-6 & 2-x \end{bmatrix} \begin{bmatrix} 1 & 0 & -y-x\left(\frac{3xy-9y}{5-3x^2}\right) & x-x\left(\frac{1-3x^2}{5-3x^2}\right) \\ 0 & 1 & \frac{3xy-9y}{5-3x^2} & \frac{1-3x^2}{5-3x^2} \\ 0 & 0 & 3+2y-B+2x & x^2-B \\ 0 & 0 & \frac{y+x-20}{16} & 1 \end{bmatrix}$$

$$\frac{y+x-20}{16} = -\frac{1}{2}$$

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Question 2

R_4

$x = 5y \qquad x - 5y = 0 \rightarrow x_1 - 5x_2 + 0x_3 + 0x_4 = 0$

Part A

if we make a equation like $\rightarrow x_1 - 5x_2 + 0x_3 + 0x_4 = 0$ the intersection with the first equation will give a volume and it will be 4D space.

$$\begin{bmatrix} 1 & -5 & 0 & 0 \\ 1 & -5 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow$$ 3 Free variables will make a volume in 4D space.

Part B

$$\begin{bmatrix} 1 & -5 & 0 & 0 & 0 \\ a & b & c & d & e \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ f \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -5 & 0 & 0 & 0 \\ 0 & b+5a & c & d & e \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ f \end{bmatrix}$$

the rank of the matrix is 2 and the free variables of matrix is 3 if $\boxed{b+5a \neq -5}$ and if we use this matrix we can have a volume in intersection

Part C

$$\begin{bmatrix} 1 & -5 & 0 \\ 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix} \quad r_2 - r_1$$

$$\begin{bmatrix} 1 & -5 & 0 \\ 0 & 9 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix} \quad r_1 + 5/9\, r_2$$

$$\begin{bmatrix} 1 & 0 & -5/9 \\ 0 & 9 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10/3 \\ 6 \end{bmatrix} \quad 1/9\, r_2$$

$$\begin{bmatrix} 1 & 0 & -5/9 \\ 0 & 1 & 1/9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10/3 \\ 2/3 \end{bmatrix}$$

$x = 10/3 + 5/9 z$

$y = 2/3 - 1/9 z$

$$x_5 = \begin{bmatrix} 10/3 \\ 2/3 \\ 0 \end{bmatrix} + z \begin{bmatrix} 5/9 \\ -1/9 \\ 1 \end{bmatrix}$$

These vectors form a vector space

These vectors span a plane in 3D vector space

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Question 3

$\vec{u} = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$ $(n \times 1)$ Let $n=3$ $\vec{u} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Part a

$H = I - 2uu^T$

$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

so $H^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = I$ //

Part b

→ $H^T \cdot H = I$ if H matrix is orthonormal

$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \neq I$ H matrix isn't orthonormal

Part c

$\lambda I - H = \begin{bmatrix} \lambda - I & 0 & 0 \\ 0 & \lambda - 1 & 0 \\ 0 & 0 & \lambda + 1 \end{bmatrix}$

$(\lambda - 1) \cdot (\lambda - 1)(\lambda + 1) = 0$

$\lambda_1 = 1$

$\lambda_2 = +1$

$\lambda_3 = -1$

for $\lambda_1, \lambda_2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} x_1 = 0$

$x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Corresponding eigenvalue → $\lambda = -1$

for $\lambda_3 \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} x_2 = 0$

$x_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Part D

Let $u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and H will be $I - \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ //

$(1-\lambda)(-1-\lambda) = 0$

$\lambda_1 = 1 \rightarrow \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \cdot x_1 = 0$

$\lambda_2 = -1 \rightarrow \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} x_2 = 0$

$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$S \cdot H \cdot S^{-1} = D$ H isn't making the equation set true

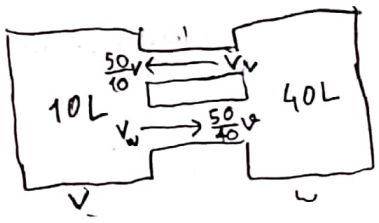
$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

So H isn't diagonal

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Question 4

$v(0)=10\,L$
$w(0)=40\,L$

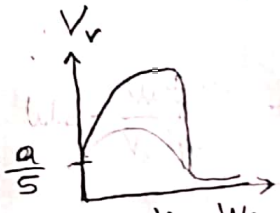


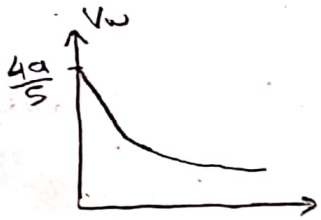
$V_v = \frac{\text{Total tank level}}{\text{tank level of w}}$

$V_w = \frac{\text{Total tank level}}{\text{tank level of v}}$

v_n : water level in tank v in the nth time interval
w_n : water level in tank w in the nth time interval

$\frac{V_{v_0}}{V_{w_0}} = \frac{10}{40}$





$\text{Volume of w} = W_0 + \frac{V_0 - W_0}{V_0 + W_0}$

$\text{Volume of v} = V_0 - \frac{V_0 - W_0}{V_0 + W_0} =$

$V_{n+1} = V_n + \frac{W_n}{V_n+W_n} - \frac{V_n}{V_n+W_n}$

$W_{n+1} = W_n + \frac{V_n}{V_n+W_n} - \frac{W_n}{V_n+W_n}$

Part B

First → %80 of tank w full
%20 of tank v full

t=1 sec there will be 3V flow to tank v

Part C

in the steady state 25 Liters will exist in both tanks.

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